

CSC 648 (Section 3)

Vibe Check: Milestone 2

ARKHA (Team 6)

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<https://github.com/sfsu-joseo/csc648-848-project-sp26-ARKHA>

Revision	Description
v3	2/27 @ 5:22pm: Milestone 2, Version 1 Draft
v2	2/24 @ 9:03 pm: Finished Milestone 1, checkpoint 3 requirements
v1	Created document 2/7 @ 5:22pm, added tech stack for Care Companion and Vibe Check apps

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Data Definitions

User

Definition: A user is a person who uses the application to view nearby locations and interact with content. A User can create Notes and can reply to or react to existing Notes.

Attribute	Description
user_id	Unique identifier for each User.
username	Public display name shown on Notes and the User Profile.
email	Email address used for login and system notifications.
password_hash	Securely hashed password. Never stored in plaintext.
profile_picture_url	URL pointing to the user's profile photo.
created_at	Timestamp of when the account was created.
account_status	Current state of the account: active, suspended, or deleted.

Location

Definition: A Location represents a physical place shown on the map (e.g., café, library, venue, park). Notes are tied to a specific Location to ensure that updates are relevant to that place.

Attribute	Description
location_id	Unique identifier for each Location.
name	Display name of the location (e.g., J. Paul Leonard Library, Philz Coffee).
address	Human-readable street address used for display and search.
latitude	GPS latitude coordinate. Used for map rendering and radius enforcement.
longitude	GPS longitude coordinate. Used for map rendering and radius enforcement.
category_tags	One or more category labels assigned to this location (e.g., Restaurant, Study Spot, Venue, Park).
avg_vibe_score	Rolling average of Vibe Scores from non-expired Notes at this location.

Note

Definition: A Note is a short, location-locked post created by a User to describe what is happening at a Location in real time. Each Note has a timestamp and an expiration time so that outdated information is automatically removed.

Attribute	Description
note_id	Unique identifier for each Note.
user_id	The User who posted the Note. Links to the User entity.
location_id	The Location this Note describes. Links to the Location entity.
content	The text of the Note. Maximum 280 characters.
vibe_score	The atmosphere rating selected by the user: Dead, Quiet, Moderate, Busy, or Buzzing.
is_anonymous	Whether the poster's username is hidden. Default is false.
created_at	Timestamp of when the Note was posted.
expires_at	Timestamp of when the Note auto-deletes. Typically 2–4 hours after created_at.

Category Tag

A Category Tag is a label used to classify a Note or Location (e.g., restaurant, study spot, venue line, parking). Category Tags allow users to filter and view relevant content more efficiently.

Attribute	Description
tag_id	Unique identifier for each Category Tag.
tag_name	Display label for the tag (e.g., Restaurant, Study Spot, Venue, Parking, Park).

Reply

A Reply is a response to a specific Note that provides additional context or updated information. Each Reply belongs to one Note.

Attribute	Description
reply_id	Unique identifier for each Reply.
note_id	The Note this Reply belongs to.
user_id	The User who posted the Reply.

content	Text content of the Reply. Maximum 280 characters.
created_at	Timestamp of when the Reply was posted.

Reaction

A Reaction represents a quick feedback action (such as an upvote or similar interaction) that a User can apply to a Note. Reactions help indicate which notes are useful or accurate.

Attribute	Description
reaction_id	Unique identifier for each Reaction.
note_id	The Note being upvoted.
user_id	The User who submitted the Reaction.
created_at	Timestamp of when the Reaction was submitted.

Heatmap Activity

Heatmap Activity is a visual representation of recent activity in a geographic area. It is generated from recent Notes and user interactions within a specific time window and helps users quickly assess how active an area is.

Attribute	Description
latitude	Center GPS latitude of the heatmap area cell.
longitude	Center GPS longitude of the heatmap area cell.
activity_score	Count of non-expired Notes in this area within the active time window. Higher score = more intense heat color.
time_window	The rolling time range used to compute activity. Default is 2 hours.
computed_at	When this heatmap snapshot was last calculated.

User Profile

A user profile stores user preferences such as notification settings, default radius, preferred categories. It tracks user contribution stats (for future gamification).

Attribute	Description
profile_id	Unique identifier for each User Profile.
user_id	The User this Profile belongs to. Each User has exactly one Profile.

default_radius_km	The default search and filter radius in kilometers. Default is 1.0 km.
preferred_categories	List of category tags the user prefers to see by default on the map.
visibility	Controls who can see this profile: public or friends only.
total_notes_posted	Lifetime count of Notes posted by this user. Used for badge tracking.
total_reactions_received	Lifetime count of upvotes received on the user's Notes. Used for badge tracking.
friend_notifications_enabled	Whether the user receives notifications when a friend leaves a nearby Note. Default is true.
notifications_enabled	Whether the user receives general system notifications. Default is true.

Notification

A notification represents a system notification (trending location nearby, etc.)

Attribute	Description
notification_id	Unique identifier for each Notification.

user_id	The User receiving this Notification.
type	The category of event: trending_location, friend_note, or badge_earned.
message	Human-readable notification text shown to the user.
is_read	Whether the user has viewed this notification. Default is false.
created_at	Timestamp of when the Notification was generated.

Friend

A Friend Connection represents a mutual relationship between two Users. It allows Users to see each other's Notes on the map and receive notifications when a friend leaves a Note nearby.

Attribute	Description
friendship_id	Unique identifier for each Friend Connection.
user_id_1	One side of the friendship.
user_id_2	Other side of the friendship.
status	Current state of the connection: pending, accepted, or blocked.

created_at	Timestamp of when the friend request was sent.
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Vibe Score

A Vibe Score is a quick, standardized rating a User can apply to a Location through their Note, similar to Google Maps' thumbs system but framed around atmosphere. Instead of stars, users rate using descriptors (e.g., Quiet, Moderate, Buzzing) or a simple scale.

Attribute	Description
vibe_level	The descriptive label selected by the user: Dead, Quiet, Moderate, Busy, or Buzzing.
numeric_value	Internal numeric mapping used for aggregate calculations: Dead=1, Quiet=2, Moderate=3, Busy=4, Buzzing=5.
note_id	The Note this Vibe Score belongs to. Each Note has exactly one Vibe Score.

Badge

A Badge is an achievement awarded to a User for meeting specific contribution milestones (e.g., leaving 10 notes, visiting 5 cafes, being a top contributor in an area).

Attribute	Description
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badge_id	Unique identifier for each Badge.
badge_name	Display name of the badge (e.g., Coffee Connoisseur, Local Legend).
description	What the user needs to do to earn this badge.
requirement_threshold	The count required to earn this badge (e.g., 10 notes at coffee shops).

Functional Requirements

Priority 1

Authentication

- FR 1.1: The user shall be able to create an account with an email and password.
- FR 1.2: The user shall be able to log in and log out.
- FR 1.3: The user shall be required to be authenticated in order to create notes, replies, or reactions.

Location

- FR 2.1: The user shall be able to grant permission for the app to access their current location.
- FR 2.2: The user shall be able to view nearby locations on the map.
- FR 2.3: The user shall be able to select a specific location from the map.

Heatmap

- FR 3.1: The user shall be able to view a heatmap showing activity in the nearby area.
- FR 3.3: The user shall be able to zoom in and out of the map.

Notes

- FR 4.1: The user shall be able to view notes related to a specific location.
- FR 4.9: The user shall be able to post a note at a specific location.
- FR 4.10: The user shall be able to compose text content when creating a note.
- FR 4.13: The user shall not be able to post from outside the defined radius
- FR 4.14: The user shall be shown an error message if posting fails.
- FR 4.15: The user shall be able to view notes in a card format resembling a letterboxed review, showing vibe level, note content, timestamp, reaction count, and reply count.
- FR 4.17: The user shall be able to view the username or anonymous handle associated with each note.
- FR 4.18: The user shall be able to view the time elapsed since a note was posted.

Vibe Score

- FR 7.1: The user shall be able to select a vibe level (Dead, Quiet, Moderate, Busy, Buzzing) when creating a note.
- FR 7.2: The user shall be able to view the aggregate vibe score of a location based only on non-expired notes.
- FR 7.3: The user shall see a fallback message if insufficient data exists for the vibe score.
- FR 7.4: The user shall be able to view vibe scores visually using descriptive labels rather than numerical star ratings.
- FR 7.5: The user shall be able to view an updated vibe score for a location when new notes are added or expired notes are removed.

Priority 2

Heatmap

- FR 3.2: The user shall be able to filter the heatmap by category.

Notes

- FR 4.2: The user shall be able to view replies to a note.
- FR 4.3: The user shall be able to reply to notes posted by other users.
- FR 4.4: The user shall be able to delete their own reply.
- FR 4.5: The user shall be able to view reactions to a note.
- FR 4.6: The user shall be able to react to notes posted by other users.
- FR 4.7: The user shall not be able to react multiple times to the same note.
- FR 4.8: The user shall be able to remove a reaction.

- FR 4.11: The user shall be able to edit their own note.
- FR 4.12: The user shall be able to delete their own note.
- FR 4.16: The user shall be able to visually distinguish between notes posted by friends and notes posted by other users on the map.
- FR 4.19: The user shall be able to view the remaining time before a note expires.

Search/Filter

- FR 6.1: The user shall be able to search for locations by name or category.
- FR 6.2: The user shall be able to filter search results to a radius.
- FR 6.3: The user shall be able to filter search results by vibe level.

Vibe Score

- FR 7.6: The user shall be able to filter locations on the map by vibe level.

Profile Management

- FR 10.1: The user shall be able to edit their username.
- FR 10.2: The user shall be notified if the username they want to change to is unavailable.
- FR 10.3: The user shall be able to change their password.
- FR 10.6: The user shall be able to view their posting history.

Moderation

- FR 11.1: The user shall be able to report a note for inappropriate content.
- FR 11.2: The user shall be able to provide a reason for reporting a note.
- FR 11.3: The user shall be able to report a reply for inappropriate content.
- FR 11.4: The user shall be able to provide a reason for reporting a reply.

Priority 3

Notification

- FR 5.1: The user shall be able to receive a notification when a trendy location is nearby.
- FR 5.2: The user shall be able to enable or disable notifications in settings.

Friends

- FR 8.1: The user shall be able to send a friend request to another registered user.

- FR 8.2: The user shall be able to accept or decline an incoming friend request.
- FR 8.3: The user shall be able to remove a friend from their friends list.
- FR 8.4: The user shall be able to view a list of their current friends.
- FR 8.5: The user shall be able to see notes left by their friends on the map, visually distinguished from notes by non-friends.
- FR 8.6: The user shall be notified when a friend leaves a note at a nearby location.
- FR 8.7: The user shall be able to enable or disable friend activity notifications in settings.
- FR 8.8: The user shall be able to block another user, preventing them from viewing their notes or sending friend requests.
- FR 8.9: The user shall not be able to view notes posted by users they have blocked.
- FR 8.10: The user shall be able to unblock a previously blocked user.
- FR 8.11: The user shall be able to see an indicator icon on map pins where a friend has recently left a note.

Badge

- FR 9.1: The user shall be automatically awarded badges when they meet defined contribution thresholds.
- FR 9.2: The user shall be able to view all badges they have earned on their profile.
- FR 9.3: The user shall be able to view all available badges and the requirements to earn them.
- FR 9.4: The user shall be notified when they earn a new badge.
- FR 9.5: The user shall be able to track the number of notes they have posted toward badge thresholds.
- FR 9.6: The user shall be able to track the number of unique location types (e.g., cafes, venues, parks) they have posted notes at.
- FR 9.7: The user shall be able to view the number of reactions their notes have received toward badge thresholds.
- FR 9.8: The user shall be able to display earned badges on their public profile.

Profile Management

- FR 10.4: The user shall be able to upload a profile picture.
- FR 10.5: The user shall be able to delete their account.

Mockups and Storyboards

System Design

Database Architecture

Backend Architecture

Network and Deployment Design

High Level APIs and Algorithms

Key Project Risks

Project Management

Team Contributions

- Harry Kakadiya (Team Lead)
 - Version 1 Contributions:
- Kaitlyn Ashby (GitHub Master, Technical Writer)
 - Version 1 Contributions:
- Rahul Srinath (Software Architect)
 - Version 1 Contributions:
- Amulya Sagi (Backend Lead)
 - Version 1 Contributions:
- Aljhay Soriano (Scrum Master)
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Member	Score
Harry	
Kaitlyn	
Rahul	
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Aljhay	